

Fresnel Biprism — I

B.Sc. (V Semester) — Physical Optics · Kumaon University

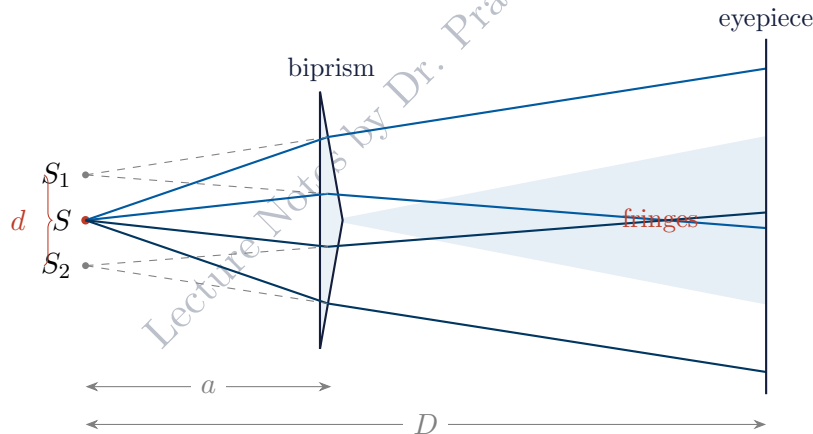
Unit II: Interference · Lecture 4 · Instructor **Dr. Prateek Sharma**

Construction · Deviation by a thin prism · Separation of virtual sources · Fringe width

1 The Fresnel Biprism

A **Fresnel biprism** is a single piece of glass whose cross-section is an obtuse-angled thin prism with an apex angle close to 180° ; equivalently, it is two thin prisms, each of small refracting angle α , joined base-to-base along the central line.

Light from a narrow slit S falls on the biprism. The upper half bends the light downward and the lower half bends it upward, so the emergent light appears to come from two *virtual* sources S_1 and S_2 . Since both are images of the same slit, they are coherent, and interference fringes are observed in the region where the two beams overlap. This is interference by *division of wavefront*.



(The angle of the biprism and the separation S_1S_2 are exaggerated. The shaded region is where the two beams overlap.)

2 Angular Deviation of Each Half

$$\delta = (\mu - 1)\alpha$$

For a thin prism of refracting angle α , applying Snell's law at both faces in the small-angle limit,

$$i_1 = \mu r_1, \quad i_2 = \mu r_2.$$

Prism geometry gives

$$r_1 + r_2 = \alpha.$$

The total deviation is

$$\delta = (i_1 - r_1) + (i_2 - r_2) = i_1 + i_2 - \alpha$$

$$\delta = \mu(r_1 + r_2) - \alpha = (\mu - 1)\alpha.$$

Note: The deviation is independent of the angle of incidence to first order — that is the defining property of a thin prism.

3 Separation of the Virtual Sources

$$d = 2a(\mu - 1)\alpha$$

Let a be the slit-to-biprism distance. Every ray through the upper half is turned through δ , so its apparent origin is shifted laterally by

$$\text{shift} = a \tan \delta \approx a\delta.$$

The lower half shifts its virtual source by the same amount on the opposite side. Hence

$$d = S_1 S_2 = 2a\delta = 2a(\mu - 1)\alpha.$$

Problem. A biprism of refractive index $\mu = 1.5$ has refracting angle $\alpha = 20'$ and is placed at $a = 2$ cm from the slit. Find the separation of the virtual sources.

Solution.

$$\alpha = 20' = \frac{20}{60} \times \frac{\pi}{180} = 5.8 \times 10^{-3} \text{ rad}$$

$$d = 2a(\mu - 1)\alpha = 2 \times 2 \times 0.5 \times (5.8 \times 10^{-3}) \text{ cm}$$

$$d \approx 0.012 \text{ cm}$$

4 Fringe Width

For a point P at a height y on the eyepiece plane, the path difference (with $d \ll D$) is

$$\Delta = S_2 P - S_1 P \approx \frac{yd}{D}.$$

Bright fringes occur where

$$\Delta = n\lambda \quad (n = 0, 1, 2, \dots)$$

$$y_n = \frac{n\lambda D}{d}$$

The fringe width is the spacing between consecutive maxima:

$$\beta = y_{n+1} - y_n = \frac{\lambda D}{d}$$

$$\beta = \frac{\lambda D}{d} \Rightarrow \lambda = \frac{\beta d}{D}$$

The wavelength can therefore be evaluated from three measurable quantities: the fringe width β (read with the micrometer eyepiece), the source separation d , and the source-to-eyepiece distance D . Measuring d is the subject of the next lecture.